

AKSHAJ MURHEKAR

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EDUCATION

The University of Texas at Austin

Aug. 2024 – May 2026

Master of Science in Information Science/Studies

GPA: 3.7/4.0

Specialization: Applied Machine Learning and Deep Learning

Selected Graduate Coursework: Deep Learning, Predictive Machine Learning, Accelerating Science with AI, Natural Language Processing, Statistical ML.

The University of Texas at Arlington

Aug. 2020 – May 2024

Bachelor of Science in Computer Science

GPA: 3.55/4.0

Selected Coursework: Artificial Intelligence, Operating Systems, Algorithms & Data Structures, Linear Algebra, Secure Programming.

PUBLICATIONS & PRESENTATIONS

Peer-Reviewed Publications

A. Murhekar, A. Mishra. *SYNAPSE: Neuro-Symbolic Visual Thought-to-Text Decoding via Topological Semantic Denoising*. Findings of **EMNLP 2026**. [PDF](#)

A. Murhekar, C. Liu, A. Mishra, S. Roychowdhury, J. Gwizdka. *SENSE: Efficient EEG-to-Text via Privacy-Preserving Semantic Retrieval*. Proceedings of **ICMI 2026**. **Oral Presentation**. [PDF](#)
Recipient of ACM SIGCHI Gary Marsden Travel Award.

Manuscripts & Presentations

S. Shukla, J. Torres, **A. Murhekar**, C. Liu, A. Mishra, J. Gwizdka, S. Roychowdhury. *A Survey on Bridging EEG Signals and Generative AI*. Arxiv 2025. [PDF](#)

[Under review for Neural Networks]

A. Murhekar, R. Tiwari. *Development of an Autonomous Medical Supply Cart to Streamline Nurse Workflow*. 2023 Biomedical Engineering Society (BMES) Annual Meeting, Seattle, WA. [Link](#)
[Poster Presentation]

RESEARCH & PROFESSIONAL EXPERIENCE

UT Austin, Research Assistant

Sep. 2026 – Present

- Conducting research with Dr. Matthew Lease on robustness in AI-assisted scientific review pipelines.

Voila Voice, AI Research Engineer

Aug. 2026 – Present

UT Austin, Graduate Research Assistant

Aug. 2025 – May 2026

- **Thesis:** Developed privacy-preserving EEG-to-text decoding methods using semantic retrieval, lightweight neural encoders, and LLM-based generation.
- Advisors: Abhijit Mishra, Shounak Roychowdhury, Jacek Gwizdka.

UTA Research Institute (UTARI), Undergraduate Research Assistant

Sep. 2023 – May 2024

- Engineered data post-processing pipelines for autonomous drone systems utilized in large-scale 3D terrain mapping and aerial reconnaissance.

UT Arlington, NSF REU, Research Fellow

Sep. 2023 – May 2024

- Architected an automated text-to-video generative pipeline for synthetic educational content featuring synchronized voiceover and visual assets.

TEACHING EXPERIENCE

UT Austin, *Graduate Teaching Assistant*

Spring 2025, Fall 2025

- **MKT 360: Information and Analysis**
- **UGS 303: The Savvy Healthcare Consumer**

UT Arlington, *Undergraduate Academic Tutor*

2022 – 2024

- **Courses Tutored:** Data Structures & Algorithms, Artificial Intelligence, Operating Systems, Discrete Mathematics, Linear Algebra, and Calculus I–III.

SELECTED RESEARCH PROJECTS

Scientific Sycophancy Benchmark: Engineered an evaluation pipeline uncovering an 85% failure rate where frontier LLMs prioritize helpfulness over physical laws, motivating a verification-aware RAG framework.

KV Cache Optimization: Engineered a PyTorch-based algorithm to reduce Llama-3 system memory usage by 50% during inference.

Audio-Visual Source Separation: Architected a U-Net with visual conditioning to separate target audio sources from mixtures via latent space modulation.

LLM Fine-tuning (Review Feedback): Fine-tuned Qwen2.5-1.5B using PEFT and Unsloth on a 46M-token synthetic dataset for high-throughput inference.

AWARDS & HONORS

ACM SIGCHI Gary Marsden Travel Award

2026

Awarded in support of presenting *SENSE: Efficient EEG-to-Text via Privacy-Preserving Semantic Retrieval* as an oral presentation at ICMI 2026.

Presidential Scholarship — Full-Tuition Merit Award (\$108,000)

2020 – 2024

A premier university-wide merit scholarship awarded to the top percentile of incoming students; provides comprehensive coverage of tuition and expenses for four years of undergraduate study.

NIH Research Grant — National Institutes of Health

2023 – 2024

Selected for NIH-funded development of autonomous healthcare robotics to optimize clinical nursing workflows.

NSF REU Award (2x Recipient) — National Science Foundation

2022 – 2024

Two-time recipient of the nationally competitive Research Experience for Undergraduates award.

2023: Funded for the design and implementation of an AI-based educational platform.

2022: Funded for the development and field-testing of an autonomous research-grade drone.

Undergraduate Research Appreciation Award — UTARI

2024

Recognized by the UTA Research Institute for technical excellence during a multi-semester research appointment.

Undergraduate Research Rank 3 — College of Engineering

2023

Ranked 3rd out of 33 competing REU teams for technical innovation and presentation of an autonomous drone system.

TECHNICAL SKILLS

Languages: Python, C/C++, SQL

ML/NLP: PyTorch, Hugging Face Transformers, PEFT/LoRA, Unsloth, vLLM, scikit-learn, RAG, instruction tuning

Systems: Linux, Git, Docker, AWS, Kubernetes, Nsight Compute